

For this deep dive I chose to focus more on the anxiety aspect, specifically what causes phobias and how best people can deal with them. The reason for this focus is because I want to find the reasoning behind my own sense of phobias (the main 2 being heights and deep water) and why I don't feel equally scared of them.

To set a base definition of phobia we will go with “A phobia is an uncontrollable, irrational, and lasting fear of a certain object, situation, or activity.”¹ As seen in the definition, a phobia is something that creates a sense of fear in you just from being around something or doing a certain activity. There isn't a singular phobia that everybody will have, but phobias are very dependent on genetic and environmental factors.

There is some research done into genetics and if phobias are passed down through families, while this is an older study, the finding from this paper does show that at least in some form anxiety disorders/phobias ². For the sake of clarity, chromosomes are what holds our genetic details and phenotypes how those genetic details are expressed. The research shows that there are certain links between a specific inclusion of a part of a chromosome and certain phobias. It also showed that if certain parts of the genome were removed, the likelihood of a phobia being shown can raise/fall. Another thing the study talks about is how just because certain indicators are present, that doesn't mean the phobia will develop/show right away. That is probably because phobias can be part of latent phenotypes In this study about phenotypes, the researchers described phenotypes as being more like circuits instead of on and off

¹ “Phobias.” *Johns Hopkins Medicine*, www.hopkinsmedicine.org/health/conditions-and-diseases/phobias.

² Villafuerte, Sandra, and Margit Burmeister. “Untangling genetic networks of panic, phobia, fear and anxiety.” *Genome biology* vol. 4,8 (2003): 224. doi:10.1186/gb-2003-4-8-224

switches³. They are all available at the same time, but the correct amount of stimulus is needed for the phenotype to be expressed, so that's why we don't always see every phenotype being expressed as one. So this shows the idea of latent phenotypes is that certain characteristics are hidden until the right environment/stimulus is applied might be the reason why we only realize we have a phobia when we encounter something. It just wasn't the right amount/correct stimulus to cause the phobia, so our brain didn't even take the time to express it. While this is a good explanation of some phobias, I don't really see how it could help explain my phobia of height and deep water.

This leads to the idea that environment/cultural factors can lead to phobias⁴. I feel that this is the more commonly expected and accepted answer to phobias. The most common example is fear of dogs. An environmental example is if you were bit by a dog, depending on if you were injured/how injured you got, a person might gain a phobia about dogs despite that being a one time event. It's more about how traumatic the event was and that lasting impression that is now imprinted on you. For a cultural example of fear of dogs can be seen in Islam. While there is nothing inherently wrong with having a dog as a pet, the saliva of dogs is seen as unclean and you have to wash yourself properly, especially before you are able to pray. This idea that dogs are unclean lead many Muslim people to be scared/nervous around dogs, not because of a bad experience, but because you would be seen as unclean/tainted if licked. In both cases the phobia is caused by factors not directly linked to your DNA, but from what you experienced while living your daily life.

³ Payne, Joshua L, and Andreas Wagner. "Latent phenotypes pervade gene regulatory circuits." *BMC systems biology* vol. 8 64. 30 May. 2014, doi:10.1186/1752-0509-8-64

⁴ "Causes of Phobias." *Cognitive Behavioral Therapy Los Angeles*, cogbtherapy.com/causes-of-phobias.

Now that brings me to a better understanding of phobias in general. It seems that phobias are a mix of genetic factors within our chromosomes and expressed by our phenotypes given the right environment. While our fear might be partially genetic based, the experiences we live play as equal to an effect on phobias we have. Negative stimuli like pain can cause us to fear something, but so could cultural/societal norms cause us to fear being ostracized or being seen as lesser/not normal. I imagine that my fear of water is partly genetic because of how easily we could die in it, but there was probably some point in my life where I had a negative experience being in water (could be something as simple as when I was being bathed as a child). I imagine there is something similar to height where being up high increases my chance of death by falling, so my body wants me to stay away from high locations.

References

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